```
        printf(" | %s\n", isHit ? "HIT" : "FAULT");
    }

    // Calculate Percentages
    float hit_percent = ((float)hits / n) * 100;
    float fault_percent = ((float)faults / n) * 100;

    printf("-----------------------------------\n");
    printf("Total Page Hits  : %d (%.2f%%)\n", hits, hit_percent);
    printf("Total Page Faults: %d (%.2f%%)\n", faults, fault_percent);

    return 0;
}
```

OUTPUT

Output

```
Optimal Page Replacement Simulation
-----------------------------------
Ref | Memory Status | Result
-----------------------------------
 1  | 1 - -  | FAULT
 2  | 1 2 -  | FAULT
 3  | 1 2 3  | FAULT
 4  | 1 2 4  | FAULT
 1  | 1 2 4  | HIT
 2  | 1 2 4  | HIT
 5  | 1 2 5  | FAULT
 1  | 1 2 5  | HIT
 2  | 1 2 5  | HIT
 3  | 3 2 5  | FAULT
 4  | 4 2 5  | FAULT
 5  | 4 2 5  | HIT
-----------------------------------
Total Page Hits  : 5 (41.67%)
Total Page Faults: 7 (58.33%)


=== Code Execution Successful ===
```